

Advanced Functions & Equations



Function composition

- 1 Let $f(x) = 3x - 2$ and $g(x) = x^2 + 1$. Find $f(g(2))$ and $g(f(2))$.
 - 2 Using the same f and g , find the formula for $(f \circ g)(x)$.
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Inverse functions

- 3 Find $f^{-1}(x)$ for $f(x) = 3x - 2$.
 - 4 Verify your answer above by computing $f(f^{-1}(5))$.
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Rational equations

- 5 Solve $\frac{2x - 1}{x + 3} = \frac{1}{2}$.
 - 6 Solve $\frac{3}{x - 1} + 2 = 5$.
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Factoring higher-degree polynomials

- 7 Given that $x = 1$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$, use the Factor Theorem to factor completely and find all roots.
 - 8 Verify that $x = -2$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$ by direct substitution.
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Exponential and logarithmic equations

- 9 Solve $3^{x-1} = 81$.
- 10 Solve $\log_3(x + 2) + \log_3(x - 2) = 2$, checking for extraneous roots.